

0 6 . 1

Describe the induced-fit model of enzyme action.

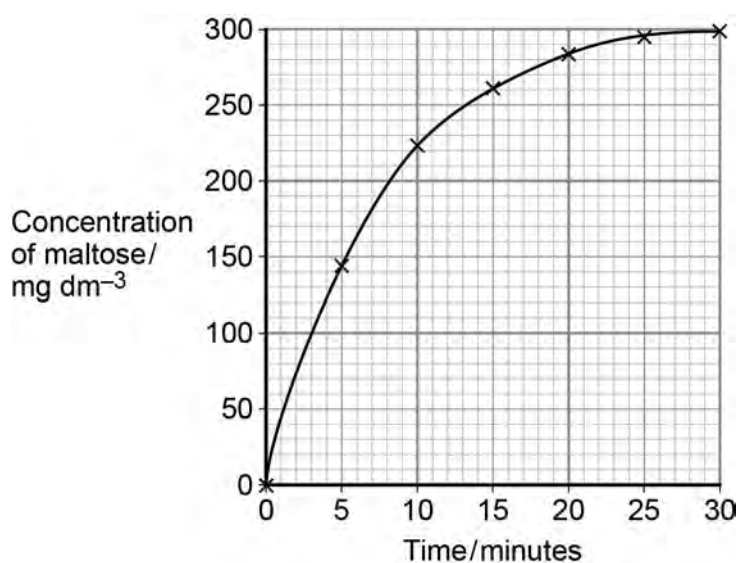
[2 marks]

0 6 . 2

A scientist investigated the hydrolysis of starch. He added amylase to a suspension of starch and measured the concentration of maltose in the reaction mixture at regular intervals.

His results are shown in **Figure 4**.

Figure 4



Determine the rate of the reaction **at** 10 minutes.
Show how you obtained your answer.

[2 marks]

Rate of reaction _____ $\text{mg dm}^{-3} \text{ min}^{-1}$



06 . 3

Explain the results shown in **Figure 4**.**[2 marks]**

06 . 4

A quantitative Benedict's test produces a colour whose intensity depends on the concentration of reducing sugar in a solution. A colorimeter can be used to measure the intensity of this colour.

The scientist used quantitative Benedict's tests to produce a calibration curve of colorimeter reading against concentration of maltose.

Describe how the scientist would have produced the calibration curve and used it to obtain the results in **Figure 4**.

Do **not** include details of how to perform a Benedict's test in your answer.

[3 marks]

